Sr.No. 8527

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:401	
	Course Section:1	Seat No: 2022600026

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Name of the candidate. Pranav Shrikant Lakhotra

Candidate's UID number: 2 0 2 2 6 0 0 0 2 6

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTech

(For example FY MCA/FY BTECH)

SEMESTER : VII

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE (AIML)

DAY AND DATE: Wednesday 8/10/25

COURSE NAME: Deep Learning.

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks | Question No. 1b]

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Q.1b 1) bias-variance decomposition for the expected squared error of a predictor is the mathematical equation of
$error^2 = bias^2 + variance +$

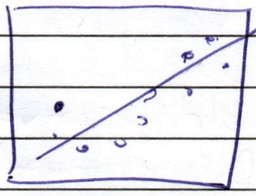
2) Overfitting is a case when variance is high & model becomes too complex.

3) It tries to learn even the unimportant features of training data due to over availability of neurons.

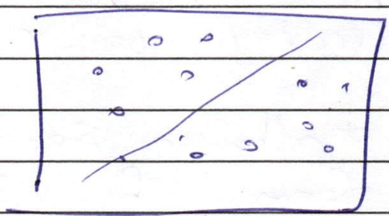
4) when a layer/model has over capacity it means that there are over neurons presents in the layers which learn from the data.

5) As they're high they try to learn irrelevant features of input data of training data.

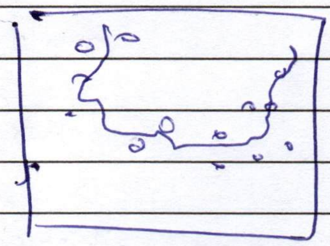
6) Since, test data is unrelated to training data, this causes these complex models to ~~fail~~ ~~at~~ give bad results with high variance.



Generalized



Underfitting



Overfitting.

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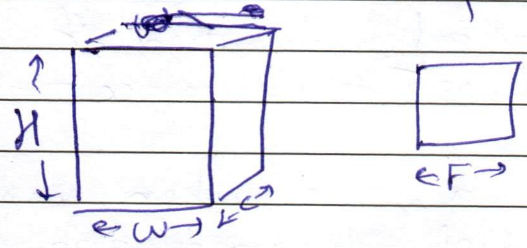
Q.2] a] 1) let the input size by $H \times W \times C$ & kernel be $F \times F$



2) Since the convolution operations occurs as the kernel is kept on input image & slided till each multiplication & calculation is done.

3) Hence the usual count of output image pixels is

$(H-F+1)(W-F+1)$ for 2D convolution.

4) Since in each pixels the number of multiplications is same as the number of pixels in kernel ~~those~~ that is $F \times F$ & ~~addition~~ & summation of values.

5) Hence, for 2D-convolution the computational complexity the values is $(H-F+1)(W-F+1) \times F \times F$ multiplication occur.

6) For depthwise seperable convolution which is a form of 3-D convolution these are just additional layers which are formed.

7) Hence for depthwise seperable convolution the number of multiplication is just the product of 2D convolution multiplications to depth C.

8) Hence, $(H-F+1)(W-F+1) \times F \times F \times C$ is the value.

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9) ~~A~~ Depthwise seperable convolution are important ~~for~~ in modern architecture as they help for 3-D designs & architecture.

10) They also help in compressing, pooling & anomaly detection for a 3D ~~str~~ structure.

11) Parameter efficiency plays a critical role for CNN as they ~~are~~ help in representing the data in concise manner as the ANN can take much more computational efficiency.

12) CNN drastically reduce the computational needs as they extract relevant data in much lesser computations.

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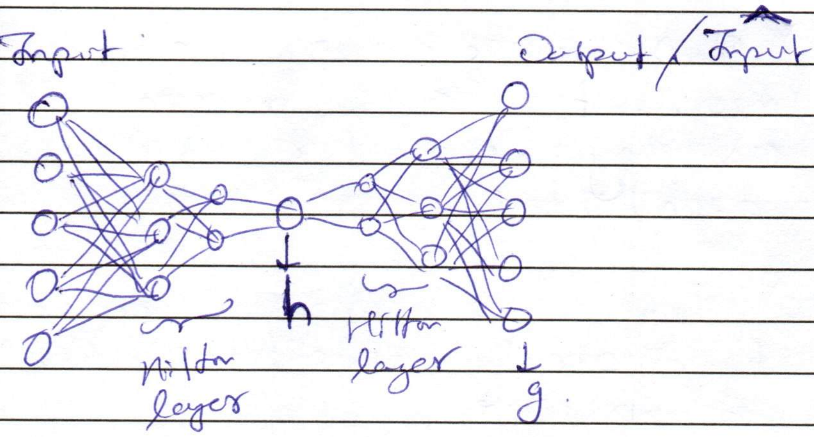
Q.2 b)

1)



2) let input be x hence the under ~~complete and~~ under complete auto encoder equation will be for encoder

$$h = f(x) = w_e x$$

3) let ~~input~~ output be $\hat{x}$ hence the decoder equation will be

$$\hat{x} = g(h) = w_d h = w_d w_e x$$

4) The MSE or cross-entropy errors of the model will be. Here MSE.

$$\sum |x - \hat{x}|^2 = |x - w_d w_e x|^2$$

5) Since the h is encoded as $h \leq x$ as this is under complete auto encoder.

6) By Mathematical representation it can be shown that $w_d w_e$ value for under complete auto encoder ~~without regularization~~ reaches the PCA equivalent.

7) Hence in this way undercomplete auto encoder linear is equivalent to PCA.

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8) Similarly for overcomplete autoencoder the capacity of $h \geq x$.

9) Since, there are no regularizations or constraints set they try to mimic the input set itself.

10) As there's no restriction on them they with free will tends to become same as input neurons.

11) Hence, as the encoder forms the same set as input for the intermediary without regularizations they're said to give identity function.

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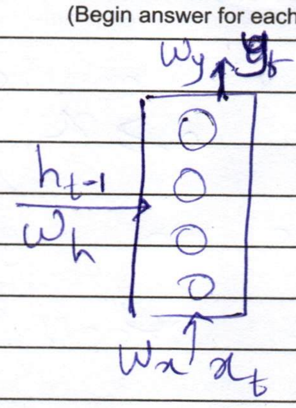
	Q.3d	

1)



2) The equations of RNN are

$$y_t = \phi(w_x x_t + w_h h_{t-1} + b_x)$$

$$L = \phi(w_y y_t + b_y)$$

Where ϕ is an activation function like ReLU or tanh.

3) replacing $y_t \rightarrow$

$$L = \phi(w_y(w_x x_t + w_h h_{t-1} + b_x) + b_y)$$

let's assume ϕ as ReLU.

4) Hence

$$L = w_x w_y x_t + w_y w_h h_{t-1} + w_y b_x + b_y$$

5) taking partial derivatives of ∂h_{t-1}

$$\boxed{\frac{\partial L}{\partial h_{t-1}} = w_y w_h}$$

6) LSTM ~~ha~~ introduces additional gates of forget, input, output & candidate

7) It tries to retain the relevant information from previous iterations into the memory for better calculations.

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8) Since it adds new gates of forget & input, it solves for vanishing gradient as now the previous info is under control.

9) But the value of how much to retain decided if vanishing gradient preserves or is solved for.

10) ~~A~~ value of $f=0$ will only take new candidate input & forget previous output. & similarly a value of $f=1$ will only keep previous info and no new input will affect.

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	Q.3b)	1) GRU ~~as~~ also tries to solve the problem of vanishing gradients over RNN.
		2) It is similar to LSTM but has lesser gates & hence lesser computation but also lesser accuracy.
		3) GRU has gates update, Reset & Candidate.
		4) $U_t = \phi(\omega_u x_t + \omega_h h_{t-1} + b_u)$
		$R_t = \omega(\omega_r x_t + \omega_h h_{t-1} + b_r)$
		where ϕ & ω are activation functions
		$\hat{C}_t = \sigma(\omega_t x_t + (R_t \odot C_{t-1})\omega + b_c)$
		$C_t = (1-Z) U_t + Z \hat{C}_t + b_t$
		5) Since, ~~as~~ the gates in GRUs are lesser then that of LSTM as LSTM has 4 Gates & GRUs have 3 Gates.
		6) Hence the computations performed by GRUs per epochs are slightly lesser then LSTM & naively can be said that they are near 25% lesser computationally expensive than LSTM.
		7) In a real world problem if we were to design a speech to text model.

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8) Using GRU models would help as fewer but more than RNN models may be required ~~as it~~

9) They can detect ~~to~~ human speech & perform further computations & help in lesser energy consumption.

10) LSTM can possibly faults to such a case or can be expensive.

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